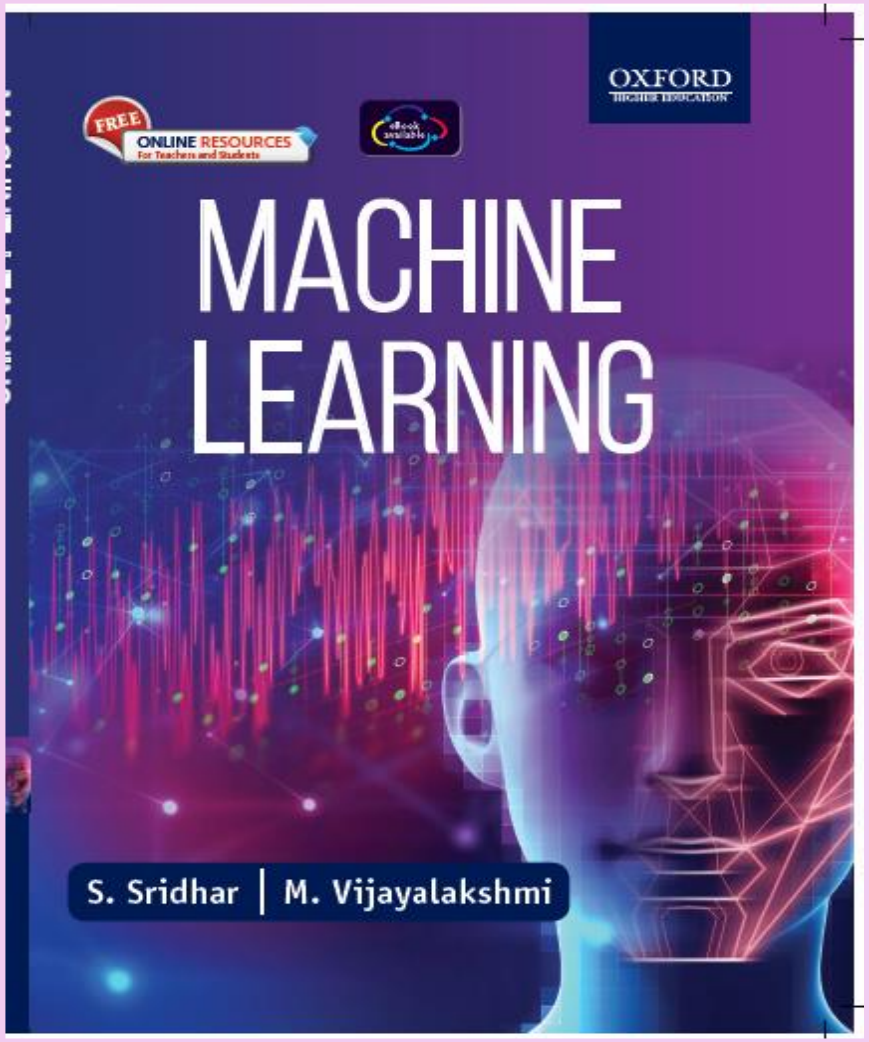


OXFORD
UNIVERSITY PRESS

Machine Learning

S. Sridhar and M. Vijayalakshmi

© Oxford University Press 2021. All rights reserved

Chapter 4

Similarity-based Learning

© Oxford University Press 2021. All rights reserved

Similarity or Instance-based Learning

Similarity-based classifiers use similarity measures to locate the nearest neighbors and classify a test instance which works in contrast with other learning mechanisms such as decision trees or neural networks. Similarity-based learning is also called as Instance-based learning/Just-in time learning since it does not build an abstract model of the training instances and performs lazy learning when classifying a new instance. This learning mechanism simply stores all data and uses it only when it needs to classify an unseen instance. The advantage of using this learning is that processing occurs only when a request to classify a new instance is given. This methodology is particularly useful when the whole dataset is not available in the beginning but collected in an incremental manner.

© Oxford University Press 2021. All rights reserved

Instance-and Model-based Learning

Table 4.1: Differences between Instance-based Learning and Model-based Learning

Instance-based Learning	Model-based Learning
Lazy Learners	Eager Learners
Processing of training instances is done only during testing phase	Processing of training instances is done during training phase

Instance-based Learning	Model-based Learning
No model is built with the training instances before it receives a test instance	Generalizes a model with the training instances before it receives a test instance
Predicts the class of the test instance directly from the training data	Predicts the class of the test instance from the model built
Slow in testing phase	Fast in testing phase
Learns by making many local approximations	Learns by creating global approximation

© Oxford University Press 2021. All rights reserved

Nearest-Neighbor Learning

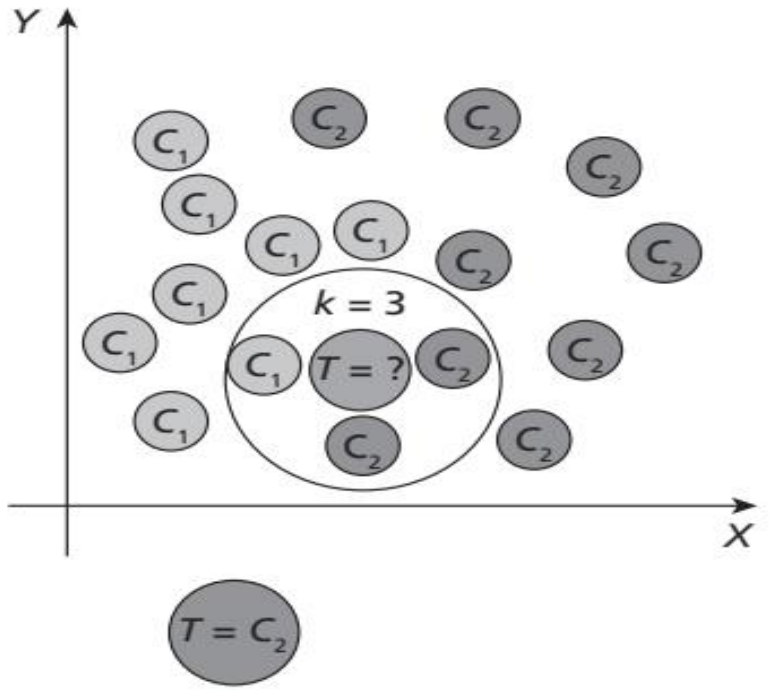


Figure 4.1: Visual Representation of *k*-Nearest Neighbor Learning

© Oxford University Press 2021. All rights reserved

Algorithm 4.1: k-NN

Algorithm 4.1: *k*-NN

Inputs: Training dataset T, distance metric d, Test instance t, the number of nearest neighbors k

Output: Predicted class or category

Prediction: For test instance t,

1. For each instance i in T, compute the distance between the test instance t and every other instance i in the training dataset using a distance metric (Euclidean distance).
   [Continuous attributes - Euclidean distance between two points in the plane with coordinates (x_1, y_1) and (x_2, y_2) is given as dist $((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$]
   [Categorical attributes (Binary) - Hamming Distance: If the value of the two instances is same, the distance d will be equal to 0 otherwise $d = 1$.]
2. Sort the distances in an ascending order and select the first k nearest training data instances to the test instance.
3. Predict the class of the test instance by majority voting (if target attribute is discrete valued) or mean (if target attribute is continuous valued) of the k selected nearest instances.

© Oxford University Press 2021. All rights reserved

Weighted k-Nearest-Neighbor Algorithm

Algorithm 4.2: Weighted *k*-NN

Inputs: Training dataset 'T', Distance metric 'd', Weighting function $w(i)$, Test instance 't', the number of nearest neighbors 'k'

Output: Predicted class or category

Prediction: For test instance t,

1. For each instance 'i' in Training dataset T, compute the distance between the test instance t and every other instance 'i' using a distance metric (Euclidean distance).
   [Continuous attributes - Euclidean distance between two points in the plane with coordinates (x_1, y_1) and (x_2, y_2) is given as dist $((x_1, y_1), (x_2, y_2)) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$]
   [Categorical attributes (Binary) - Hamming Distance: If the values of two instances are the same, the distance d will be equal to 0. Otherwise $d = 1$.]
2. Sort the distances in the ascending order and select the first 'k' nearest training data instances to the test instance.
3. Predict the class of the test instance by weighted voting technique (Weighting function $w(i)$) for the k selected nearest instances:
   - Compute the inverse of each distance of the 'k' selected nearest instances.
   - Find the sum of the inverses.
   - Compute the weight by dividing each inverse distance by the sum. (Each weight is a vote for its associated class).
   - Add the weights of the same class.
   - Predict the class by choosing the class with the maximum vote.

© Oxford University Press 2021. All rights reserved

Nearest Centroid Classifier

Algorithm 4.3: Nearest Centroid Classifier

Inputs: Training dataset *T*, Distance metric *d*, Test instance *t*

Output: Predicted class or category

1. Compute the mean/centroid of each class.
2. Compute the distance between the test instance and mean/centroid of each class (Euclidean Distance).
3. Predict the class by choosing the class with the smaller distance.

© Oxford University Press 2021. All rights reserved

Locally Weighted Regression (Lwr)

Locally Weighted Regression (LWR) is a non-parametric supervised learning algorithm that performs local regression by combining regression model with nearest neighbor's model. LWR is also referred to as a memory-based method as it requires training data while prediction but uses only the training data instances locally around the point of interest. Using nearest neighbors algorithm, we find the instances that are closest to a test instance and fit linear function to each of those '*k*' nearest instances in the local regression model. The key idea is that we need to approximate the linear functions of all '*k*' neighbors that minimize the error such that the prediction line is no more linear but rather it is a curve.

© Oxford University Press 2021. All rights reserved

Summary

1. Similarity-based learning is a supervised learning technique that predicts the class label of a test instance by measuring the similarity of this test instance with training instances.
2. Similarity-based classification problems formulate the features of test instance and training instances in Euclidean space to learn the similarity or dissimilarity between instances.
3. A natural approach to similarity-based classification is k-Nearest-Neighbors (k-NN), which is a non-parametric method used for both classification and regression problems.
4. k-NN predicts the category of the test instance according to the 'k' training samples which are closer to the test instance and classifies it to that category which has the largest category probability.
5. Data normalization/standardization is required when data (features) have different ranges or wider range of possible values when computing distances and to transform all features to a specific range.
6. k-NN best suits for lower dimensional data, as in a high-dimensional space the nearest neighbors may not be very close at all.
7. Weighted k-NN assigns a higher weight to the k closest neighbors to the test instance in the decision than neighbors that are farther away from the test instance.
8. Nearest Centroid Classifier is a simple classifier also called as Mean Difference classifier that classifies a test instance to the class whose centroid/mean is closest to that instance.
9. Locally Weighted Regression (LWR) is a non-parametric supervised learning algorithm that performs local regression by combining regression model with nearest neighbor's model.

© Oxford University Press 2021. All rights reserved